

Predictive Modeling for Course Demand and Revenue Forecasting on EduPro

Submitted By

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Project Domain: Machine Learning / Predictive Analytics

Tools Used: Python, Jupyter Notebook, VS Code, Streamlit

Libraries Used: Pandas, NumPy, Matplotlib, Scikit-learn, Joblib

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Acknowledgement

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I am thankful to my mentors and instructors for providing guidance and valuable learning resources throughout this project.

I would also like to acknowledge the use of modern AI-based learning tools such as **ChatGPT** by [OpenAI](#) and **Gemini** by [Google Gemini](#), along with online resources available through [Google Search](#), which helped me understand concepts, debug coding errors, and improve my project workflow.

These tools were used strictly for learning, guidance, problem-solving, and understanding technical concepts such as machine learning, data preprocessing, model deployment, and Streamlit development.

However, all implementation, coding decisions, analysis, testing, and final project understanding were carried out by me.

This project significantly improved my practical knowledge of machine learning, predictive analytics, and real-world problem solving.

Abstract

With the rapid growth of online education, predicting future course demand and revenue has become very important for online learning platforms. Platforms need to understand which courses are likely to perform well so they can make better business decisions regarding pricing, course launches, and instructor allocation.

This project focuses on building predictive models for **EduPro**, an online learning platform. The main aim of this project was to analyze historical data related to courses, teachers, and transactions in order to predict future course demand and revenue generation.

During this project, I performed data cleaning, feature engineering, exploratory data analysis, and machine learning model development. Models such as Linear Regression and Random Forest Regressor were used for prediction. In addition, a Streamlit dashboard was created to visualize insights and provide interactive revenue prediction.

This project demonstrates how machine learning can transform raw historical data into useful business insights and support smarter decision-making.

1. Introduction

In recent years, online learning has become one of the fastest-growing industries. Students and working professionals use online platforms to learn new skills such as programming, data science, machine learning, business analytics, and digital marketing.

As the number of available courses increases, online learning platforms face an important challenge: **predicting future demand**. It becomes difficult to answer important business questions such as:

- Which course category will attract more learners?
- Which courses will generate higher revenue?
- What pricing strategy should be followed?
- Which instructors should be assigned to premium courses?

Without proper analysis, these decisions are often based on assumptions or previous experience.

This project aims to solve this problem using machine learning. By analyzing historical EduPro data, predictive models were developed to estimate future course demand and revenue. This helps the platform make better decisions and reduce business risk.

2. Problem Statement

EduPro currently does not have a predictive system for forecasting course demand and revenue. Most business decisions are based on historical reports instead of future predictions.

Because of this, EduPro faces several challenges:

- Difficulty identifying high-demand courses
- Uncertainty in pricing strategies
- Inefficient instructor allocation
- Increased business risk during new course launches

For example, launching a course without understanding expected demand may lead to poor enrollments and low revenue. Similarly, incorrect pricing can reduce both student interest and profitability.

Therefore, there is a strong need for predictive analytics to support data-driven decision-making.

3. Objectives of the Project

The main objectives of this project were:

1. To analyze historical EduPro course data
2. To predict future course enrollment demand
3. To forecast course revenue
4. To identify major factors affecting revenue
5. To develop an interactive dashboard for visualization and prediction

The overall goal was to convert historical data into actionable business insights.

4. Dataset Description

The dataset used in this project consisted of three different sheets.

4.1 Courses Sheet

This sheet contains course-related information.

Features included:

- CourseID
- CourseName
- CourseCategory
- CourseType
- CourseLevel
- CoursePrice
- CourseDuration
- CourseRating

These features help describe course characteristics such as pricing, duration, category, and quality.

4.2 Teachers Sheet

This sheet contains instructor-related information.

Features included:

- TeacherID

- Expertise
- YearsOfExperience
- TeacherRating

These features help evaluate instructor quality and experience.

4.3 Transactions Sheet

This sheet contains purchase transaction records.

Features included:

- TransactionID
- CourseID
- TransactionDate
- Amount

This sheet was mainly used for revenue analysis.

5. Data Preparation

The first step in the project was loading and understanding the dataset.

All Excel sheets were loaded using Python and Pandas. Before building machine learning models, it was important to inspect the data and understand its structure.

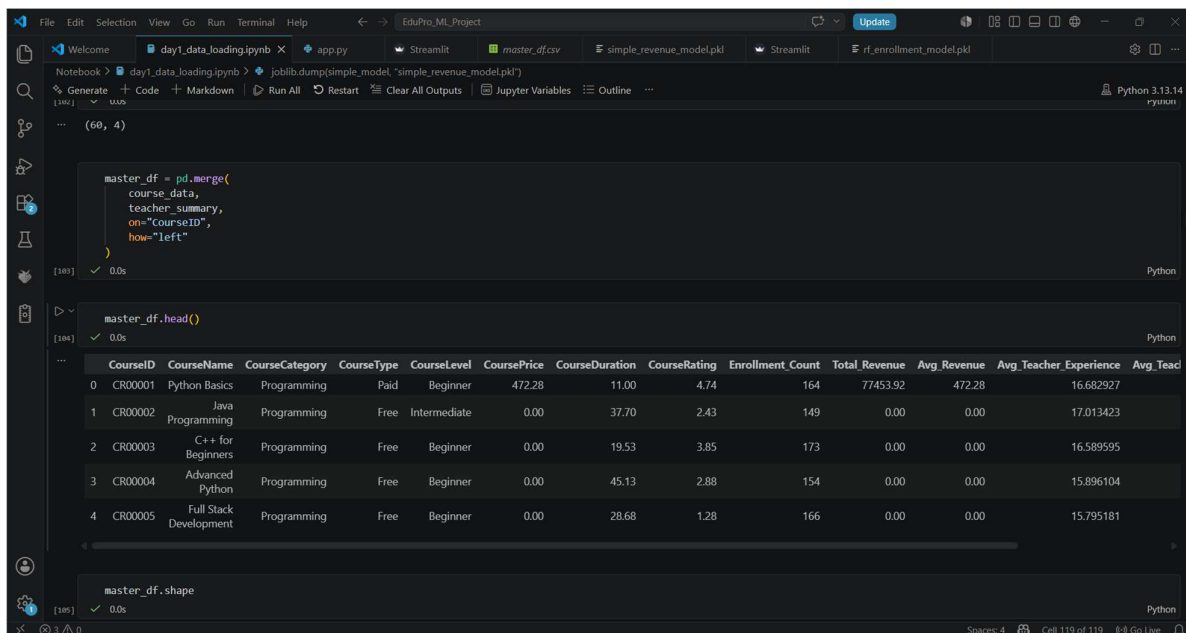


Figure 1: Dataset preview after loading and merging all sheets

The following tasks were performed:

- Loading datasets
- Checking column names
- Verifying data types
- Identifying missing values
- Understanding data relationships

The datasets were then merged to form a master dataset.

6. Data Merging

Since useful information was distributed across three sheets, the datasets needed to be combined into a single dataset.

The merging process connected:

- Course data
- Teacher data
- Transaction data

These datasets were merged using common keys such as:

- CourseID
- TeacherID

After merging, a **master dataset** was created containing all relevant features needed for analysis and prediction.

This made further preprocessing and model training easier.

7. Data Preprocessing

Data preprocessing is one of the most important steps in machine learning because raw data usually contains inconsistencies.

Several preprocessing steps were performed.

7.1 Handling Missing Values

Missing values were identified and handled properly.

Missing values can negatively affect model performance, so they were either:

- Replaced using mean values
- Cleaned manually
- Removed when necessary

This improved dataset quality.

7.2 Feature Engineering

Feature engineering was performed to create useful derived features.

New features included:

Course Features

- Price bands (Low / Medium / High)
- Duration buckets
- Rating tiers

Instructor Features

- Experience buckets
- Teacher rating score

Performance Features

- Enrollment count
- Revenue per course
- Average revenue

These engineered features helped improve predictive performance.

7.3 Encoding Categorical Variables

Machine learning models cannot directly understand text values.

Columns such as:

- CourseCategory
- CourseType
- CourseLevel

were converted into numerical format using encoding methods such as:

- Label Encoding
- One-Hot Encoding

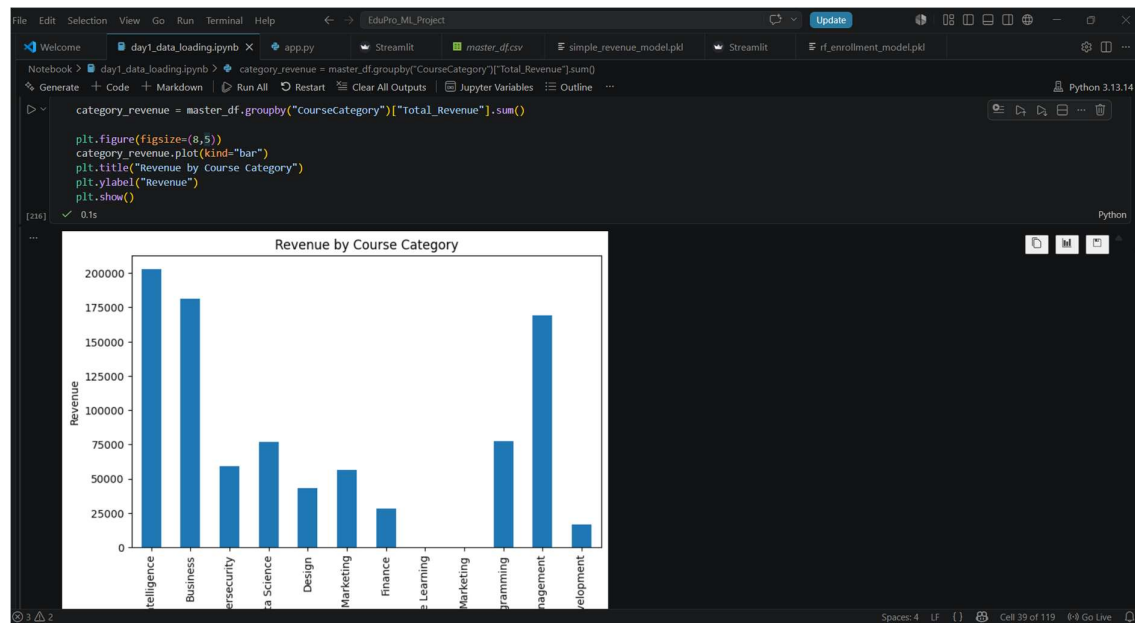
This allowed the models to process categorical data efficiently.

7.4 Feature Scaling

Numerical features were scaled to improve model performance.

Scaling ensures all features are on similar ranges and prevents large-value features from dominating smaller ones.

This step improved training efficiency.



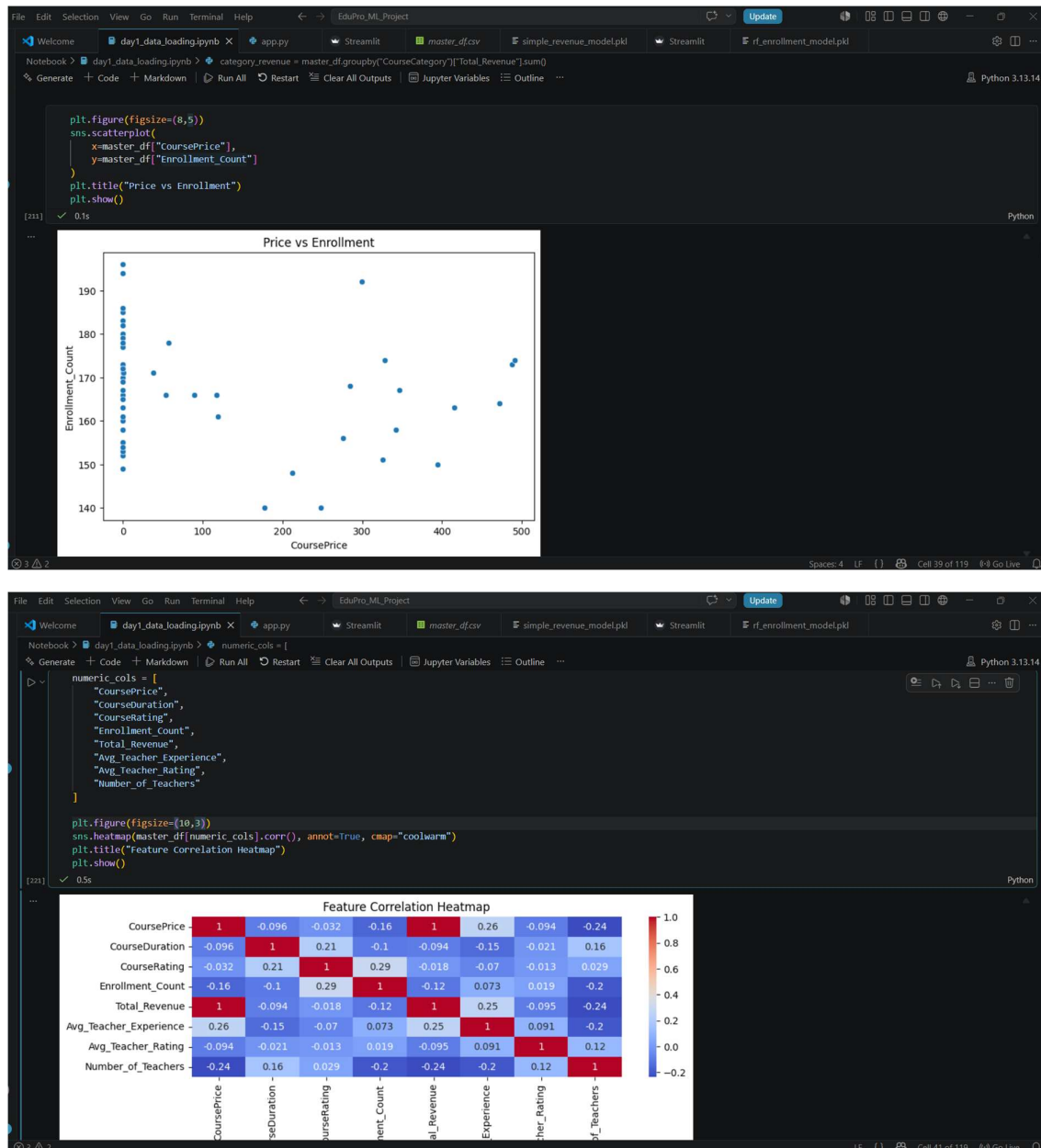


Figure 3: Revenue analysis across course categories

8. Exploratory Data Analysis (EDA)

EDA helped uncover important patterns and relationships in the data.

Several visualizations were created to better understand business trends.

8.1 Revenue by Course Category

One important analysis involved comparing revenue across different course categories.

It was observed that:

- Premium technical courses generated higher revenue
- Data Science and AI-related courses performed strongly
- Low-cost courses had higher enrollments but lower revenue

This indicates category plays an important role in revenue generation.

8.2 Course Rating vs Revenue

A relationship between course ratings and revenue was observed.

Higher-rated courses generally generated more revenue.

This suggests students prefer well-rated courses and are willing to pay more for quality content.

8.3 Price vs Demand

Course pricing significantly influenced student demand.

Observations showed:

- Extremely high prices reduced enrollments
- Reasonable pricing improved demand
- Premium pricing worked only for high-quality courses

This confirms pricing sensitivity among learners.

8.4 Teacher Experience vs Revenue

Teacher experience showed some impact on revenue, but its effect was weaker compared to pricing and course quality.

This suggests students value content quality and course relevance more than instructor experience alone.

9. Machine Learning Models Used

Two regression models were used in this project.

9.1 Linear Regression

Linear Regression was used as a baseline model.

It assumes a linear relationship between input variables and target output.

Advantages:

- Simple to implement
- Fast training
- Easy to interpret

Limitations:

- Cannot capture complex nonlinear patterns

This model provided baseline performance for comparison.

9.2 Random Forest Regressor

Random Forest Regressor was the main advanced model used in this project.

Random Forest is an ensemble learning technique that combines multiple decision trees.

Advantages include:

- Handles nonlinear relationships
- Reduces overfitting
- High prediction accuracy
- Works well with mixed features

This model performed significantly better than Linear Regression.

10. Model Evaluation Metrics

To evaluate model performance, three major metrics were used.

Mean Absolute Error (MAE)

MAE measures the average prediction error.

Lower MAE indicates better model performance.

Root Mean Squared Error (RMSE)

RMSE penalizes larger prediction errors more heavily.

Lower RMSE indicates more accurate predictions.

R² Score

R² Score measures how well the model explains variance in the data.

Interpretation:

- 0 = Poor model
- 1 = Perfect model

Higher R² indicates better predictive power.

11. Results and Findings

After training multiple models, their performance was compared.

Enrollment Prediction Results

Enrollment prediction performance was relatively weak.

This happened because enrollment depends on several external factors that were not available in the dataset, such as:

- Marketing campaigns
- Seasonal demand
- Discounts and promotions
- Competition

Because these factors were missing, enrollment prediction accuracy remained limited.

Revenue Prediction Results

Revenue prediction performed much better.

The best-performing model was:

Random Forest Regressor

This model achieved strong performance because it captured complex relationships between features such as:

- Price
- Ratings
- Category

- Duration

Revenue prediction was therefore highly reliable.

```
File Edit Selection View Go Run Terminal Help EduPro_ML_Project Update
Welcome day1_data_loading.ipynb X app.py Streamlit master_df.csv simple_revenue_model.pkl Streamlit rf_enrollment_model.pkl
Notebook > day1_data_loading.ipynb > rf_model.fit(X_train, y_train)
Generate + Code + Markdown Run All Restart Clear All Outputs Jupyter Variables Outline Python 3.13.14
[159] ✓ 0.0s
rf_mae = mean_absolute_error(y_test, rf_pred)
rf_rmse = mean_squared_error(y_test, rf_pred) ** 0.5
rf_r2 = r2_score(y_test, rf_pred)

print("Random Forest MAE:", rf_mae)
print("Random Forest RMSE:", rf_rmse)
print("Random Forest R2:", rf_r2)

Random Forest MAE: 11.908333333333337
Random Forest RMSE: 14.813803247872125
Random Forest R2: -0.20429201219512239

comparison = pd.DataFrame({
    "Model": ["Linear Regression", "Random Forest"],
    "MAE": [mae, rf_mae],
    "RMSE": [rmse, rf_rmse],
    "R2": [r2, rf_r2]
})

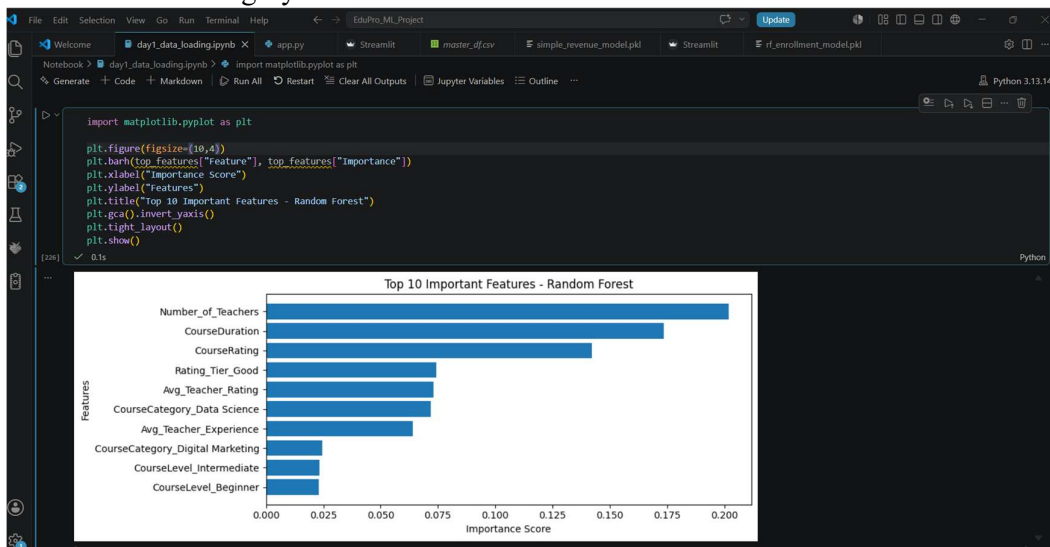
comparison
Model MAE RMSE R2
0 Linear Regression 17.177360 19.886725 -1.170327
1 Random Forest 11.908333 14.813803 -0.204292
```

12. Feature Importance Analysis

Feature importance analysis helped identify the most influential variables affecting revenue.

The most important features were:

1. Course Price
2. Course Rating
3. Course Duration
4. Course Category



Less important features included:

- Teacher Experience

This suggests that course quality and pricing play the biggest role in revenue generation.

13. Streamlit Dashboard Development

To make the project interactive and user-friendly, a dashboard was developed using Streamlit.

The dashboard includes:

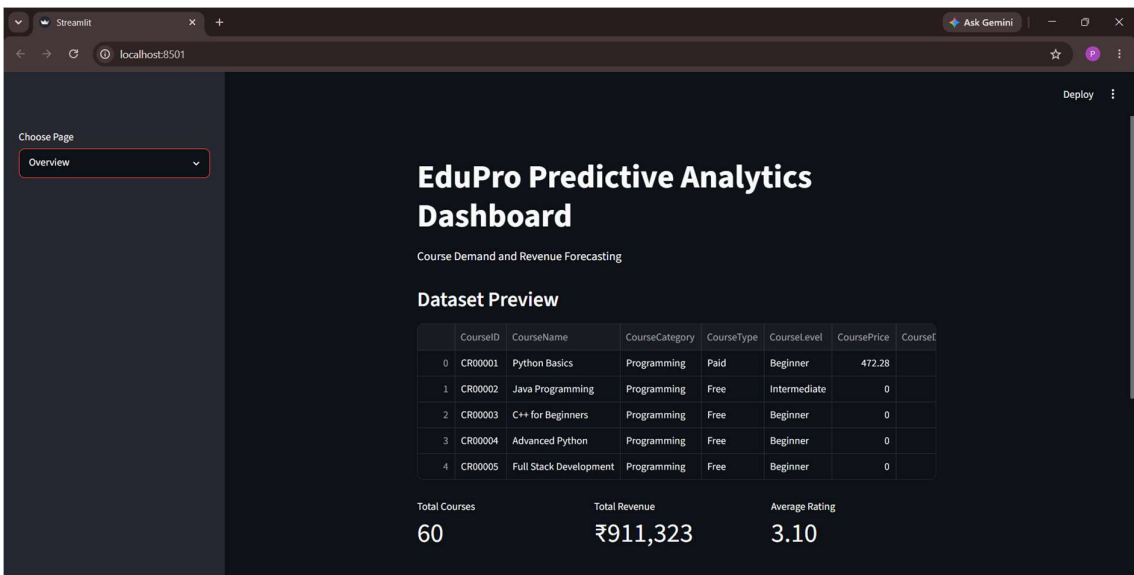
- Dataset preview
- KPI metrics
- Revenue visualizations
- Prediction interface

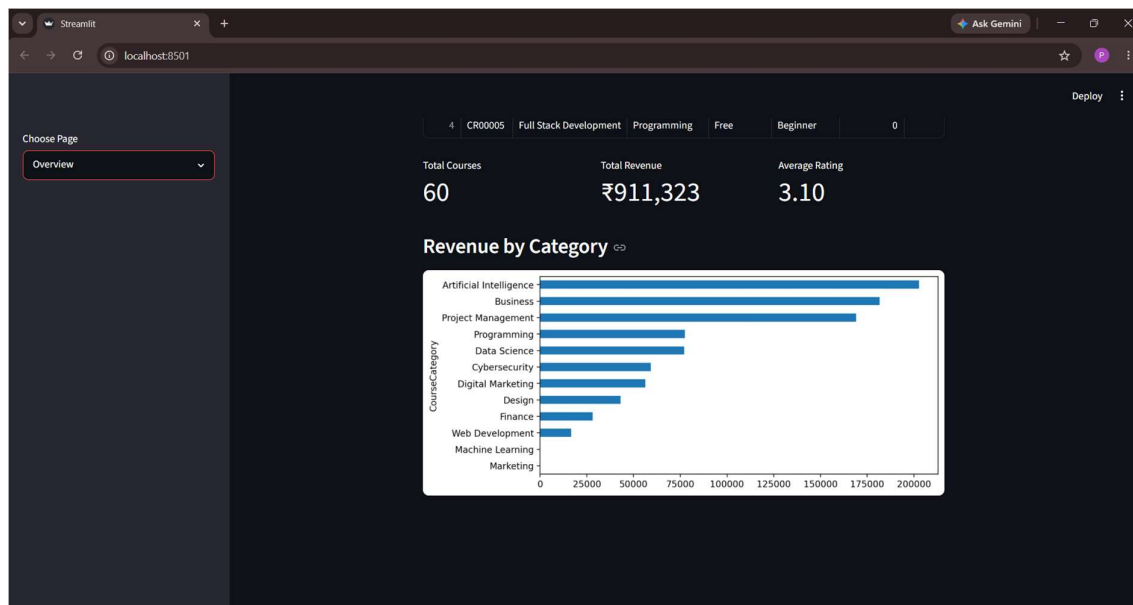
Users can input values such as:

- Course Price
- Course Duration
- Course Rating
- Teacher Experience

The dashboard then predicts expected revenue.

This makes the project more practical and business-oriented.





The screenshot shows the 'Prediction' page of the Streamlit application. The sidebar has 'Prediction' selected. The main content area is titled 'Course Demand and Revenue Forecasting' and 'Course Prediction'. It features input fields for Course Price (5000), Course Duration (30), Course Rating (4.00), and Teacher Experience (5). A 'Predict' button is located below these fields. A green message box indicates 'Prediction Complete!'. The predicted revenue is shown as ₹80,078.

Input	Value
Course Price	5000
Course Duration	30
Course Rating	4.00
Teacher Experience	5
Predicted Revenue	₹80,078

14. Challenges Faced During Project

Several challenges were encountered during this project.

Some major challenges included:

- Handling missing values
- Merging multiple datasets
- Feature engineering
- Windows file path issues

- Streamlit setup errors
- Model deployment challenges
- Feature mismatch during prediction

Each issue was solved through debugging and testing.

These challenges improved my practical understanding of machine learning workflows.

15. Business Recommendations

Based on the analysis, the following recommendations were made for EduPro.

Recommendation 1

Invest more in high-demand technical courses such as:

- Data Science
- Artificial Intelligence
- Machine Learning

These categories showed strong revenue potential.

Recommendation 2

Use optimized pricing strategies.

Avoid both:

- Overpricing
- Underpricing

Balanced pricing improves profitability.

Recommendation 3

Focus on improving course quality and ratings.

Higher-rated courses consistently generated more revenue.

Recommendation 4

Use predictive analytics before launching new courses.

This reduces uncertainty and supports better planning.

16. Conclusion

This project successfully demonstrated how machine learning can help an online learning platform make better business decisions.

Through this project, I learned how to:

- Clean and preprocess real-world datasets
- Perform feature engineering
- Build machine learning models
- Evaluate model performance
- Deploy interactive dashboards

The project showed that revenue can be predicted with good accuracy using machine learning models, especially Random Forest Regressor.

EduPro can now use predictive analytics for:

- Course planning
- Pricing decisions
- Resource allocation
- Revenue forecasting

Overall, this project helped me gain both technical knowledge and practical experience in machine learning and predictive analytics.

Future Scope

This project can be improved further by:

- Using deep learning models
- Including seasonal demand data
- Adding marketing campaign data
- Improving enrollment prediction
- Deploying dashboard to cloud servers